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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CHE 221 — Past Questions & Materials
by Renance

Course: CHE 221
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 200

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Paper 1 of 1 — CHE 221 (Introduction to Chemical Engineering)

Level 200 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN



**COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF CHEMICAL ENGINEERING**

SECOND SEMESTER; 2022/2023 SESSION EXAMINATION

COURSE TITLE: INTRODUCTION TO CHEMICAL ENGINEERING

COURSE CODE: CHE 221

Credit Units: 2 **Time Allowed: 2 Hours.**

INSTRUCTIONS: Answer any Four (4) Questions. All Questions carry equal marks

1. Differentiate between units and dimension with two examples for each
 - a. Show how these two values of R are equivalent

$$R = 0.0821 \frac{\text{L} \cdot \text{ATM}}{\text{mol} \cdot \text{K}} = 8.31 \frac{\text{J}}{\text{mol} \cdot \text{K}}$$
 - b. State the following laws (1) Raoult's law (2) Henry's law.
2. (a) 100 moles of A are fed to a batch reactor and the final product contains 10 moles of A, 160 moles of B and 10 moles of C according to the equations

$$A \rightarrow 2B \quad (\text{desired})$$

$$A \rightarrow C \quad (\text{undesired})$$

Calculate

 - (i) % yield of B (ii) Selectivity of B relative to C.
 - (b) Antimony is obtained by heating pulverized stibnite, Sb_2S_3 with scrap iron and the molten Antimony is drawn out from the bottom of the reaction

$$\text{Sb}_2\text{S}_3 + 3\text{Fe} \rightarrow 2\text{Sb} + 3\text{FeS}$$

$\downarrow$
Stibnite

$\downarrow$
Antimony

Suppose that 0.6kg of stibnite and 0.2kg of iron turnings are heated together to give 0.2kg of antimony metal. Determine

1

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- Limiting reactant
- % excess reactant
- Degree of completion
- % conversion

Given that

Component	Molecular Weight (g/mol)
Sb ₂ S ₃	339.7
Fe	55.85
Sb	121.8
FeS	87.91

- 10kg of Heptane reacts completely with the stoichiometric quantity of oxygen gas. How many kilogram of carbon (iv) oxide will be form?
- Write out the stoichiometric equation for the combustion of butane with oxygen gas
 - If 20g of butane is consumed in (bi), what mass of O₂ gas will be required for complete combustion of butane
 - What are the mass of the gases in the product?

4 a) In a drying process moisture is reduced from 60% to 30%. Initial weight of the material is 200kg. Calculate the final weight of the product.

b) Consider a continuous process in which milk continuously flows in and out. How much of concentrated milk of 35% total solid is required to produce 150 kg/h of milk powder of 95% total solid?

- Define a Unit operation
 - List 5 examples of unit operation

b) It is determined to use a continuous centrifuge to separate 35,000kg of whole milk containing 4% fat into skim milk and fat. If the skim milk is to contain 0.45% fat and the cream 45% fat after separation, determine the flow rates of these two output streams after a 6 hour period.

6. Make sketches of symbols of the following as they would appear in a process flow diagram:

Plate column Batch centrifuge Reciprocating pump Bucket elevator Atmospheric storage
Packed column Heat exchanger Rotary compressor Cyclone Belt conveyor

